PGR 210 - Natural Language Processing Part

Kristiania University of Applied Sciences

By Huamin Ren

Huamin.ren@kristiania.no

Outline

1. Similary metric
2. K-means clustering

1. Similarity metric

- Similarity scores (and distances) tell on how similar or how far apart two documents are based on the similarity (or distance) of the vectors you used to represent them
- Some familiar distances:
  - Euclidean or Cartesian distance, or root mean square error (RMSE): 2-norm or L2 , MSE
  - Squared Euclidean distance, sum of squares distance (SSD): L2
  - Cosine or angular or projected distance: normalized dot product
  - Minkowski distance: p-norm or L_p
  - Fractional distance, fractional norm: p-norm or L_p for $0 < p < 1$
  - City block, Manhattan, or taxicab distance; sum of absolute distance (SAD)

Pairwise distances available in sklearn

- 'cityblock', 'cosine', 'euclidean', 'l1', 'l2', 'manhattan', 'braycurtis', 'canberra', 'chebyshev', 'correlation', 'dice', 'hamming', 'jaccard', 'kulsinski', 'mahalanobis', 'matching', 'minkowski', 'rogerstanimoto', 'russellrao', 'seuclidean', 'sokalmichener', 'sokalsneath', 'sqeuclidean', 'yule’

1.1 Distance

- Distance measures are often computed from similarity measures (scores) and vice versa such that distances are inversely proportional to similarity scores.
- For distances and similarity scores that range between 0 and 1, like probabilities, it's more common to use a formula like this:

  similarity = 1. - distance

  distance = 1. - similarity

$$hd(u,v)=\sum_{i=1}^{n}(u_i \neq v_i)$$

$$norm\_hd(u,v)=\frac{\sum_{i=1}^{n}(u_i \neq v_i)}{n}$$

U=[1 1 1 0 0] v=[0 0 0 1 1]
What is hd? What is norm_hd?

1.2 Use similarity on?

- **Lexical similarity:** This involves observing the contents of the text documents with regards to its syntax, structure, and content and measuring their similarity based on these parameters.
- **Semantic similarity:** This involves determining the semantics, meaning, and context of the documents and then determining how close they are to each other.

- **Term similarity:** Similarity between individual tokens or words
- **Document similarity:** Similarity between entire text documents

2. K-means Clustering

- The k-means clustering algorithm is a centroid-based clustering model that tries to cluster data into groups or clusters of equal variance.
- Disadvantage of this algorithm is that the number of clusters (**k**) needs to be specified in advance.
- Advantage: perhaps the most popular clustering algorithm, due to its ease of use as well as it being scalable with large amounts of data.

Mathematical definition

- A dataset **X** with **N** data points or samples, the task is to group them into K clusters, where **K** is a user-specified parameter.
- The k-means clustering algorithm will segregate the **N** data points into **K** disjoint separate clusters, C_k, and each of these clusters can be described by the means of the cluster samples.

X = {x1,x2…xn}
c1 = {x1, x2} c2={x3}…c5…
X1=[1 0 0 1 1 1] x2=[1 0 0 1 1 0] c1 = [1 0 0 1 1 0.5]
c1={mean(x1,x2)}

- These means become the cluster centroids μ_k such that these centroids are not bound by the condition that they have to be actual data points from the **N** samples in **X**.
- [1 0] [0 1][0.5 0.5] C1: {1:[0.5 0.5]}

K-means procedure

Steps:

1. Choose initial **k** centroids *μk* by taking **k** random samples from the dataset **X**.
2. Update clusters by assigning each data point or sample to its nearest centroid point.
3. Recalculate and update clusters based on the new cluster data points for each cluster obtained from Step 2. Mathematically this can be represented as follows:

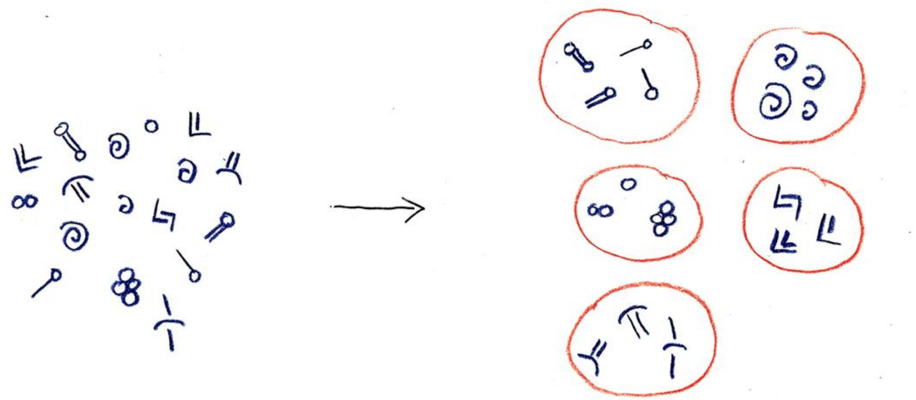
$$C_k = \{x_n : \|x_n - \mu_k\| \leq all \|x_n - \mu_l\|\}$$

$$\min \sum_{i=1}^{K} \sum_{x_n \in C_i} \| x_n - \mu_i \|^2$$

Introduction to unsupervised learning

- When we don't have labels associated with our data, we use techniques that are know collectively as *unsupervised learning*.
- Clustering, or grouping

https://www.youtube.com/watch?v=-HbHNamsVgQ
https://www.youtube.com/watch?v=dj9bl5rjqMc



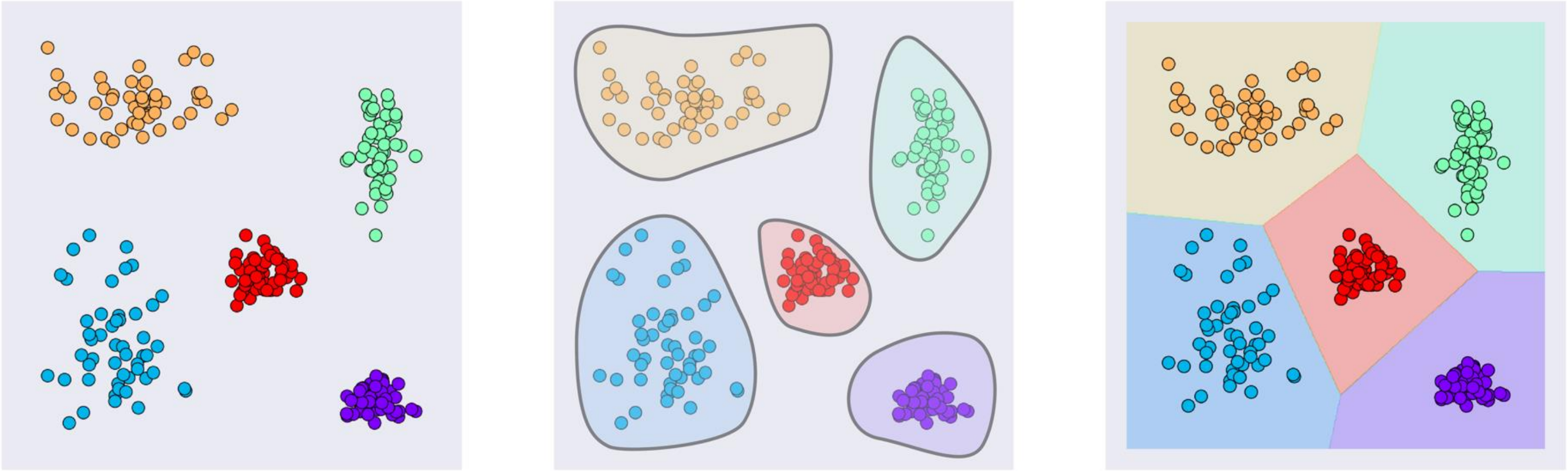
Using a clustering algorihtm to organize marks on clay pots.

Introduction to unsupervised learning

- These algorithms learn about relationships between the elements of the input, rather than between each input and a label

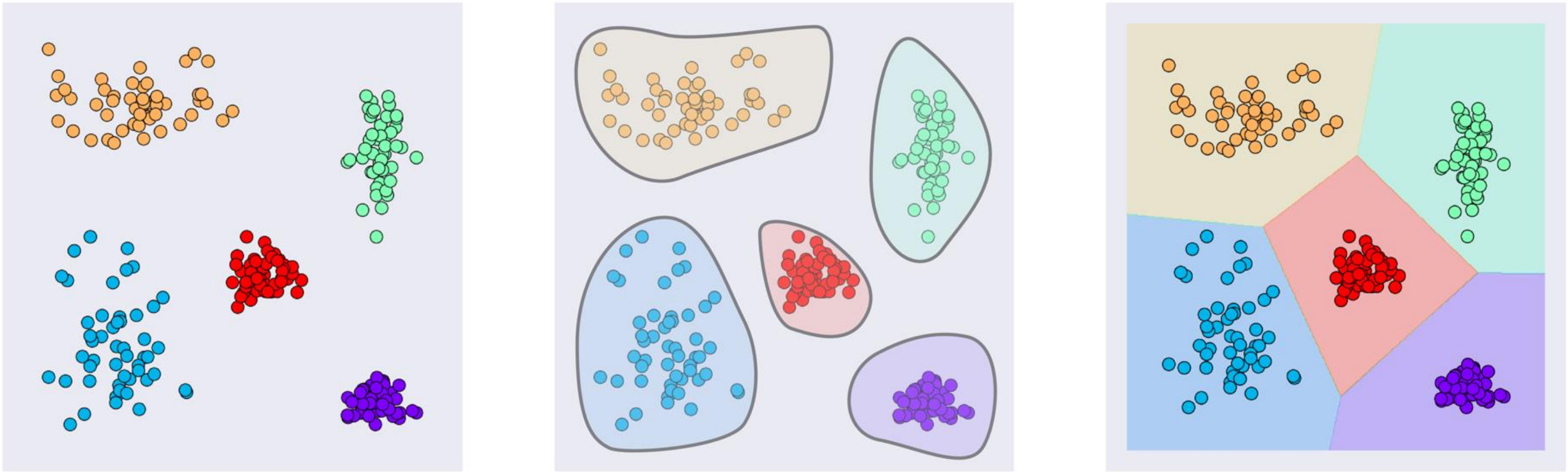
Unsupervised learning and Clustering

- What if we don’t have labels?
  - Unsupervised learning: data without labels fall into the type of learning
- K-means clustering



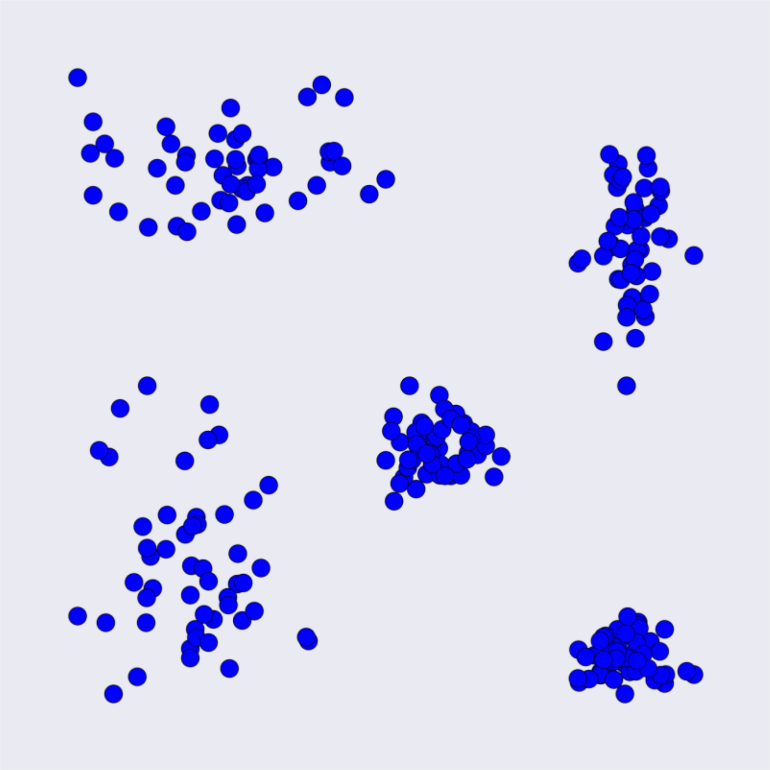
Unsupervised learning and Clustering

- K-means clustering
  - K: the number of clusters

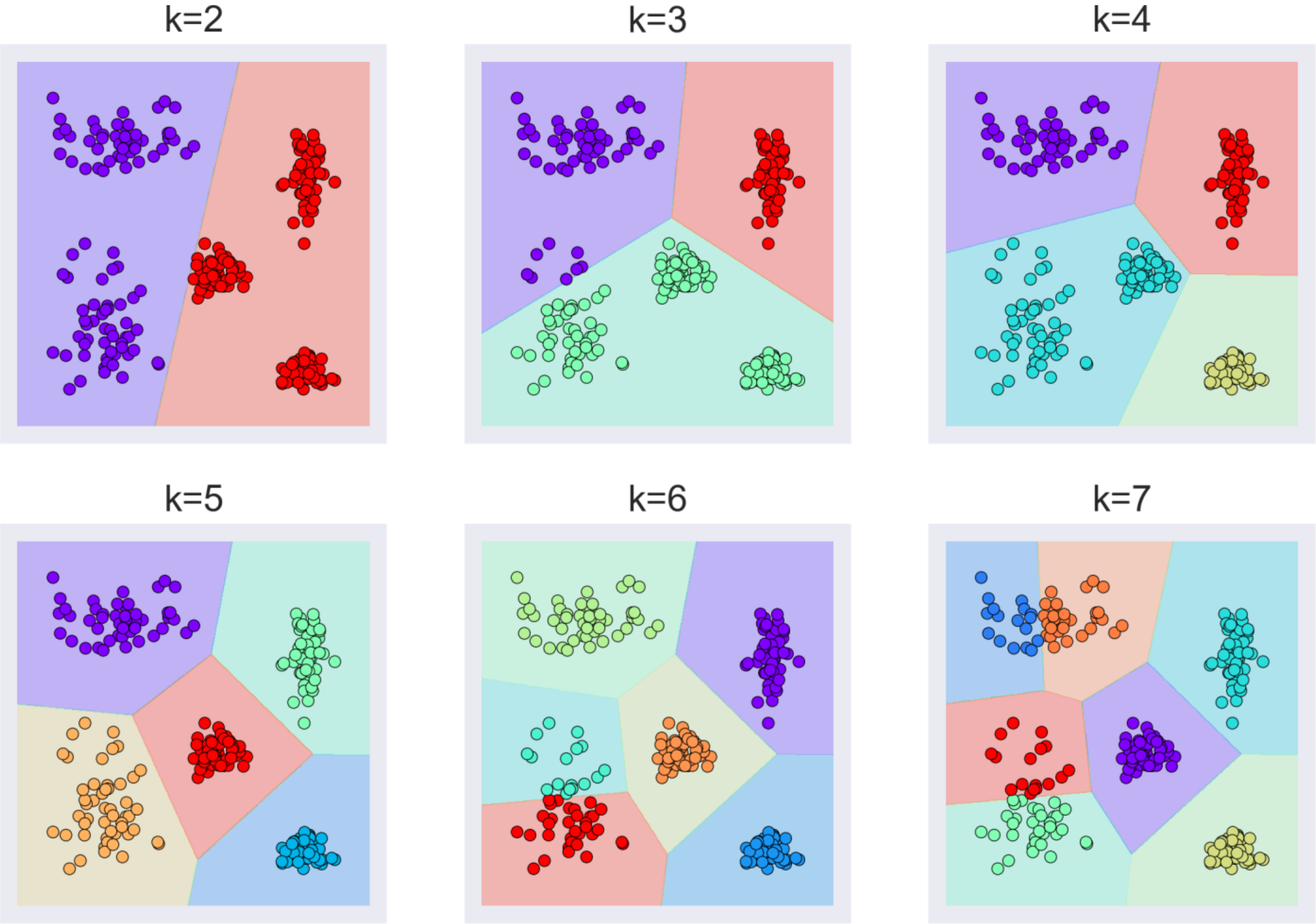


Growing clusters. Left: Starting data with five classes.
Middle: Identifying the five groups.
Right: Growing the groups outward so that every point has been assigned to one class.

K-means clustering



A set of 200 unlabeled data points.



K-means clustering

- Having the freedom to choose the value of k:
  - Upside: if we know in advance how many clusters there ought to be, we can say so, and the algorithm produces what we want
  - Downside: we may not have any idea of how many clusters best describe our data